Group Assignment

Due by 6:00 pm on October 31, 2025

Instructions

- This is a group assignment, and only one submission per group is needed.
- Please clearly indicate your group number and the members of your group in your submission.
- You're welcome to do the data exercise using R, but you're free to use other statistical software, such as Python, etc.
- Your submission can include two files: 1) A concise write-up that contains your answers. 2) A separate code file that can reproduce your results.
- Save your file names as ``[Class number]"+ ``Group * Homework," where * is your group number. Example: ``654 Group 1 Homework"

I. Fraud Detection Confusion Matrix (5 points)

Suppose a bank builds a logistic regression model to predict whether a transaction is fraudulent. The model uses the transaction amount as an independent variable to predict whether the transaction is fraudulent. The Fraud indicator represents the actual transaction is fraudulent(T) or not (F). The predicted probability of each transaction being a fraudulent transaction is listed in the last column Predicted probability.

Transaction ID	Transaction Amount	Fraud Indicator	Predicted Probability
5	$30	F	0.05
4	$50	T	0.14
8	$100	T	0.32
7	$700	F	0.48
1	$770	F	0.55
6	$990	T	0.61
3	$1200	F	0.72
2	$1300	T	0.82

1) Suppose banks use 0.4 as the cutoff to classify the predicted transaction to be fraud. Namely, the transaction is considered fraudulent if the predicted probability is greater than or equal to the cutoff 0.4. If the predicted probability is smaller than the cutoff, the transaction is deemed to be nonfraudulent.
   Please fill in the following confusion matrix. Please also calculate the True positive rate and False positive rate.

	Precited Positive	Predicted Negative
Actual Positive		
Actual Negative		

2) Suppose banks use 0.6 instead as the cutoff to classify the predicted transaction as fraud. Please repeat the above question. First, fill in the confusion matrix. Then please also calculate the True positive rate and False positive rate.

3) What tradeoff does the bank face when they decide on the cutoff? Can we compare the performances between the two cutoffs? If so, please indicate how we should compare them.

II. Daily active user and Trading volume on Robinhood (5 points)

In the following exercise you will use linear regression model to explore the relationship between the number of users traffics on Robinhood with their trading volume. The data set includes the following columns:

1) Date
2) Daily Active Users (DAU): the total number of active users on Robinhood on that date
3) Trading Volume (in shares): the total number of shares traded on the platform on that day

The dataset is a simulated dataset, and they are saved in robindhood.csv

1. Create a scatter plot to visualize the relationship between daily active users (x-axis) and trading volume (y-axis)

2. Perform a linear regression with trading volume as the outcome variable and daily active users as the independent variables.
   Hint: In R, you can use the lm() function. Then, use the summary() function to print the results.

3. What is the slope of the regression line? Interpret the slope in the context of the problem.

4. Use the model to predict the trading volume if the number of daily active users is 1,500,000.

III. Predict consumer loan defaults (10 points)

In the following exercises, you will build a credit model and use it to predict consumer loan defaults. You will start with a sample database of 10,000 three-year consumer loans - "pslcdata.csv."

This is the same dataset we used for class. The database contains the following details for each borrower: ID, Loan Amount, Interest Rate (annual in percentage), Credit Grade, Detailed Credit Grade, Length of Employment, Revolving Credit Balance, Revolving Credit Utility (in percentage), FICO score, Home Ownership Status, Annual Income, Employment Verification, Loan Status and Default status of the loan.

We will first use debt to income ratio to **predict default**.

1. Create a new variable called 'default dummy,' where 1 indicates a loan default and 0 indicates no default.

2. You also want to create debt to income ratio based on the data. Debt to income ratio can be defined as the borrower's loan amount to annual income ratio (This is often used to measure borrowers' ability to pay loans relative to their income level).
   You can name the new variable as **debt_to_income**. Draw a histogram to see how the distribution looks like for this new variable.

3. Estimate a logistic regression model with debt-to-income ratio as the predictor. Report your estimates for the intercept and the coefficient for debt to income.

   What does the coefficient tell you on the relationship between debt-to-income ratio and default probability?

4. Plot an ROC curve corresponding to the estimated model. What is the area under the curve? Is the area below your ROC curve more significant than 0.5? Report on the area.

5. Suppose you develop a policy that rejects all the loan applicants whose predicted probability of default is greater than 20%.

   1) What is the proportion of consumers you correctly predicted to default (The denominator should be the total number of people who defaulted.)

   2) What is the proportion of consumers you mistakenly predicted to default? (The denominator should be the total number of people who do not default.)

Suppose now you include additional variables in your model. You must run a logistic regression with FICO, debt to income ratio, and revolving credit balance.

6. Run a logistic regression using the explanatory variables listed above. Report your estimates for the intercept and the coefficients for all variables.

7. Plot the ROC curve. Is the area below your new ROC curve greater than that in part 3? Report on the area. Interpret the differences.

8. Suppose you develop a policy that rejects all the loan applicants whose predicted probability of default is greater than 10%.

    1) What is the proportion of consumers you correctly predicted to default (The denominator should be the total number of people who defaulted.)

    2) What is the proportion of consumers you mistakenly predicted to default? (The denominator should be the total number of people who do not default.)